



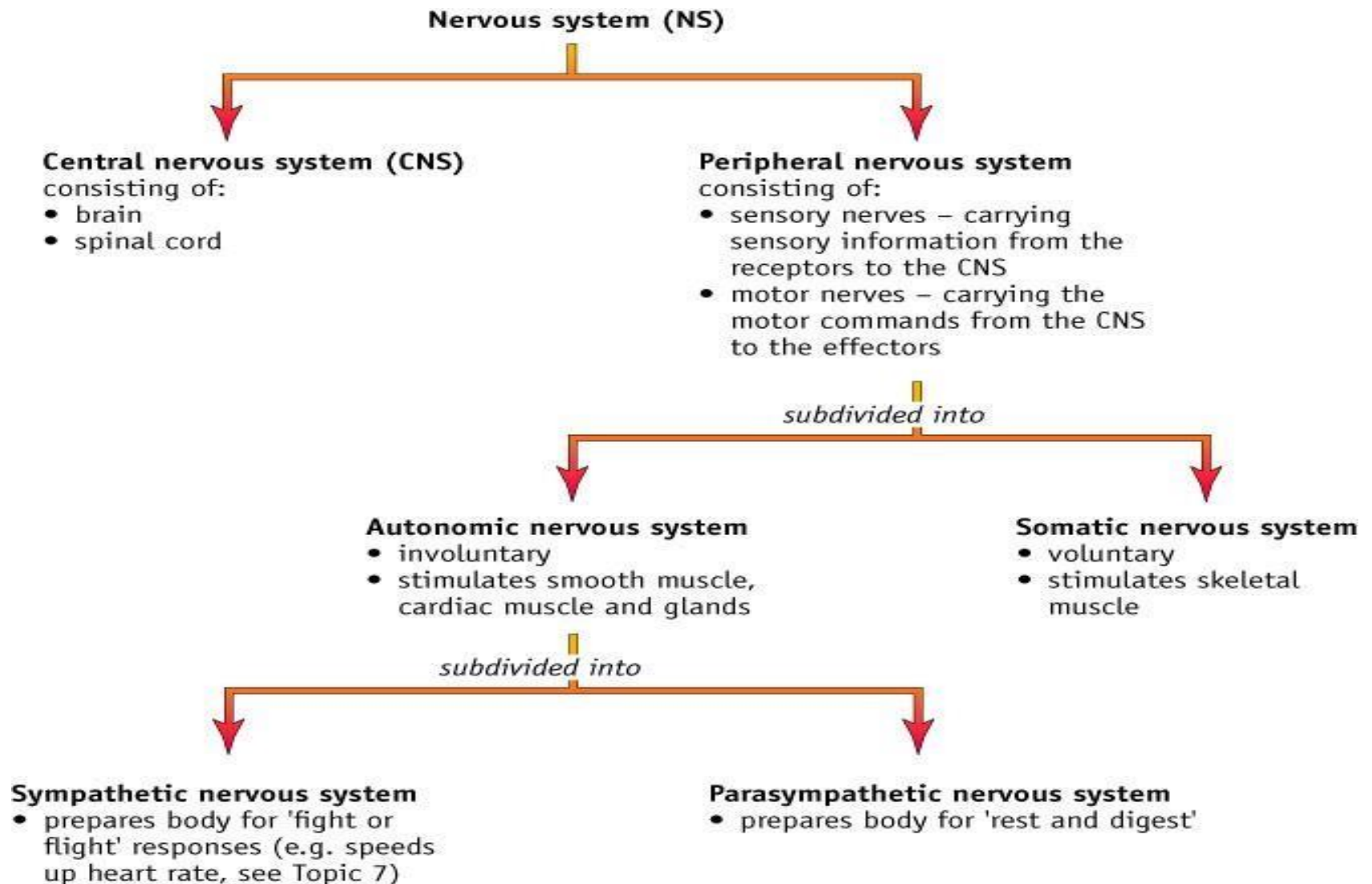
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AUTONOMIC NERVOUS SYSTEM

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Organization of Nervous system



The Autonomic Nervous System

- ❖ The peripheral nervous system consists of the **somatic nervous system** and the **autonomic nervous system (ANS)**.
- ❖ The Somatic NS consists of motor neurons that stimulate skeletal muscles.
- ❖ In contrast, the ANS consists of motor neurons that control smooth muscles, cardiac muscles, and glands.
- ❖ In addition, the ANS monitors visceral organs and blood vessels with sensory neurons, which provide input information for the CNS.

The Autonomic Nervous System

- ❖ The ANS is further divided into the sympathetic nervous system (SNS) and the parasympathetic nervous system (PNS). Both of these systems can stimulate and inhibit effectors organs
- ❖ The SNS and the PNS are continually active; impulses are transmitted at a low rate continuously through most of the nerve fibers of SNS and PNS. The basal rates of activity are known as sympathetic tone and parasympathetic tone, respectively.
- ❖ Both sympathetic and parasympathetic nervous system have preganglionic neurons that originate in CNS.
- ❖ Both have postganglionic neurons that originate outside of the CNS in ganglia

❖ Ganglia are ovoid structures containing cell bodies of neurons and glial cells supported by connective tissue. Ganglia function like relay stations - one nerve enters and another exits.

❖ The main difference between ganglia and nuclei is that ganglia are clusters of nerve cell bodies in the PNS whereas nuclei are clusters of nerve cell bodies in the CNS.

The *autonomic nervous system* controls most visceral functions of the body.

Examples :

- ☐ control arterial pressure
- ☐ gastrointestinal motility
- ☐ gastrointestinal secretion,
- ☐ urinary bladder emptying,
- ☐ sweating
- ☐ body temperature
- ☐and many other activities.

Some of these activities are controlled almost entirely and some only partially by the autonomic nervous system.

GENERAL ORGANIZATION OF THE AUTONOMIC NERVOUS SYSTEM

- ❖ ANS output can be influenced by many higher brain regions, but the main areas involved in autonomic control include the brainstem, the hypothalamus, and the limbic system.
- ❖ The ANS is activated by centers in the **spinal cord, brainstem, hypothalamus, and parts of the cerebral cortex.**

1. Spinal cord autonomic reflexes

2. Medulla

- Vasomotor center
- Respiratory center
- Swallowing, coughing, and vomiting centers

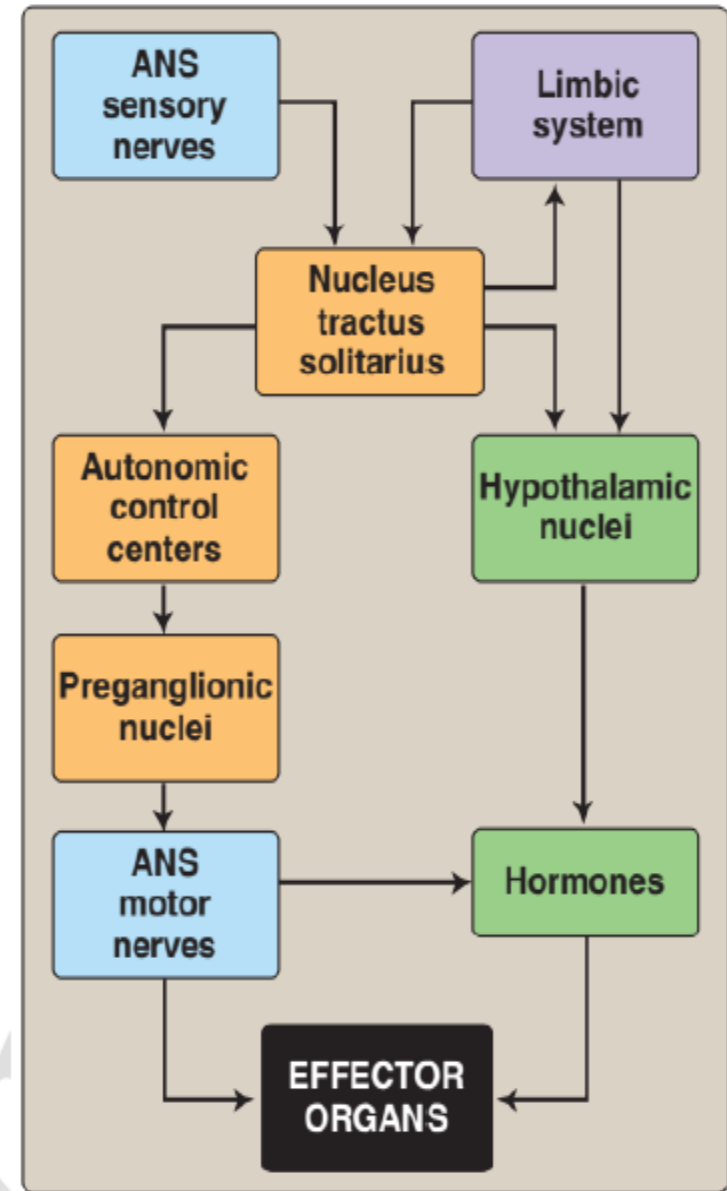
3. Pons

- Pneumotaxic center
- Micturition center

4. Hypothalamus (The main overall integration center of ANS “the boss”)

- Temperature regulation center
- Thirst and food intake regulatory centers

5. Cerebral cortex (frontal lobe, and limbic system).



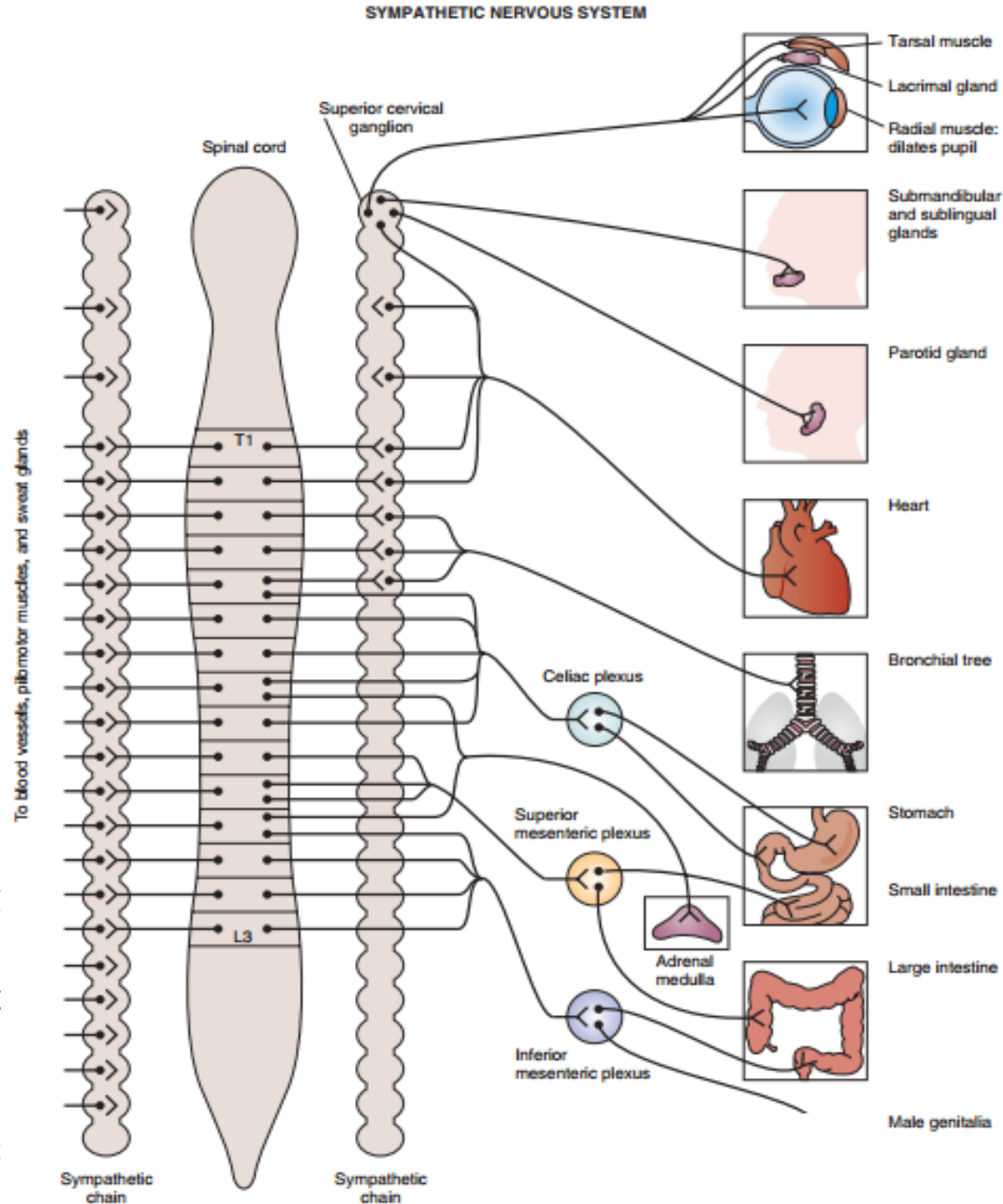
Genrally, SNS composed of the following structures

(1) *Paravertebral sympathetic chains* of ganglia that are interconnected with the spinal nerves on the side of the vertebral column,

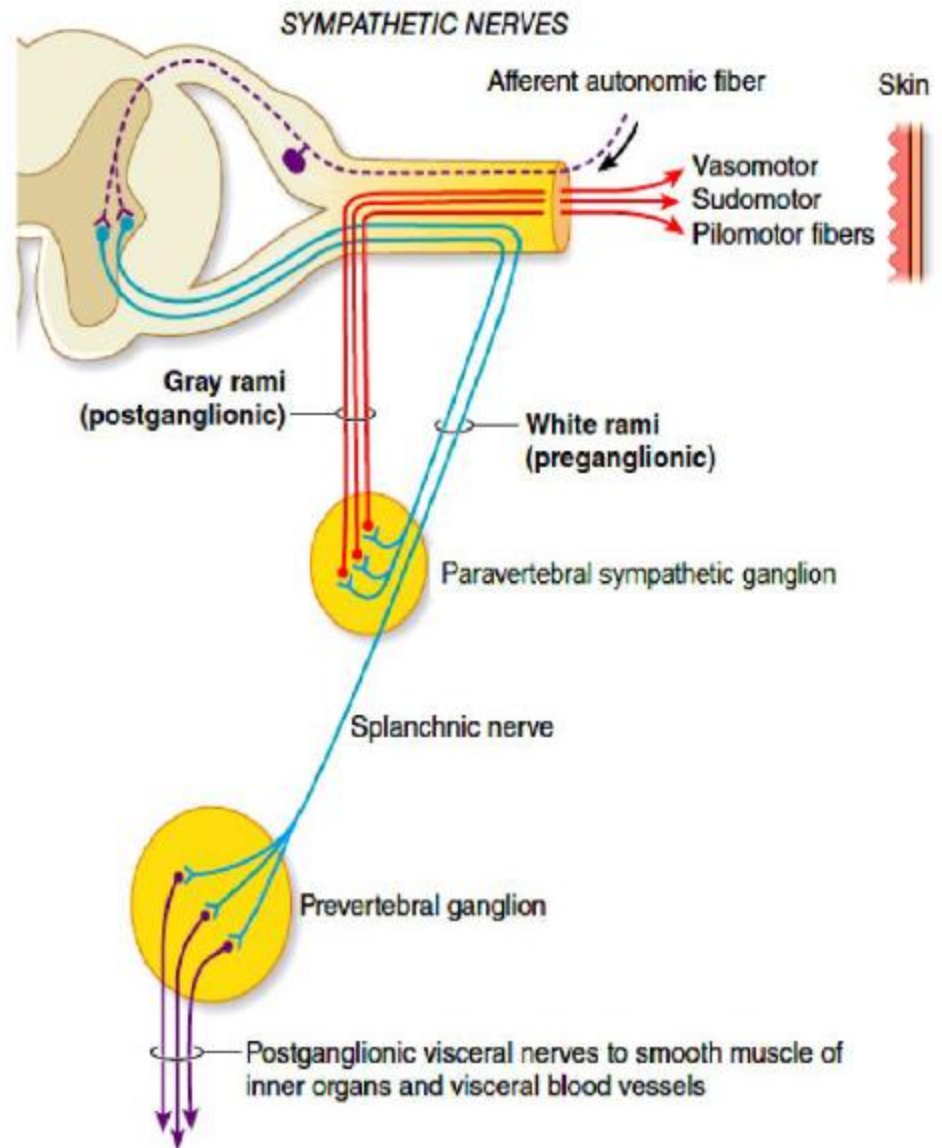
(2) *prevertebral ganglia* (the *celiac*, *superior mesenteric*, *aortico-renal*, *inferior mesenteric*, and *hypogastric*), and

(3) *nerves* extending from the ganglia to the different internal organs.

❖ The sympathetic nerve fibers originate in the spinal cord along with spinal nerves between cord segments T1 and L2 and pass first into the *sympathetic chain* and then to the tissues and organs that are stimulated by the sympathetic nerves.

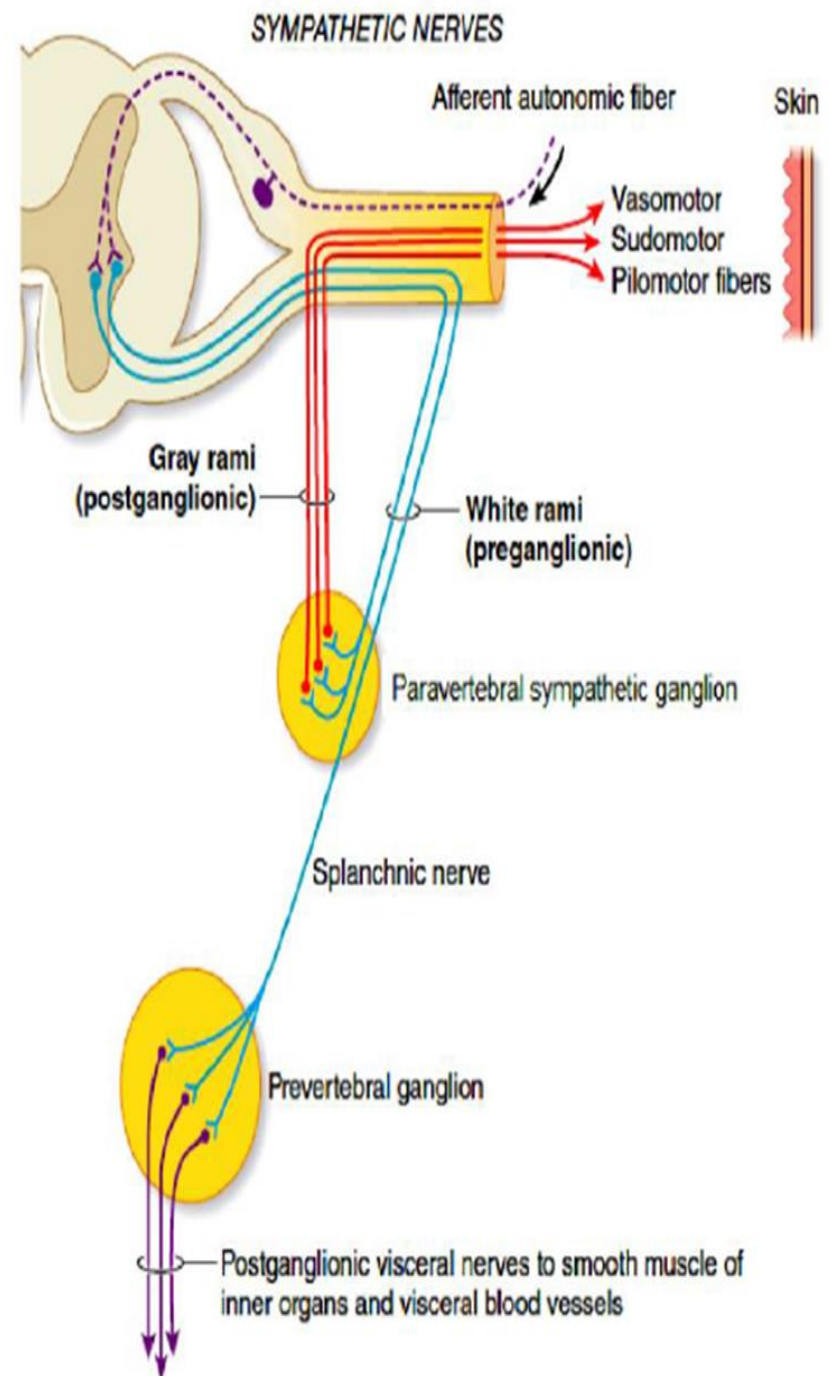


- Sympathetic division anatomy (Thoracolumbar)
- **Preganglionic neurons** between segments T1 and L2 – lateral gray horn of spinal cord
- **Preganglionic fibers**
 - Short
 - Travel in the ventral root and spinal nerve
- **Postganglionic fibers**
 - Long fibers



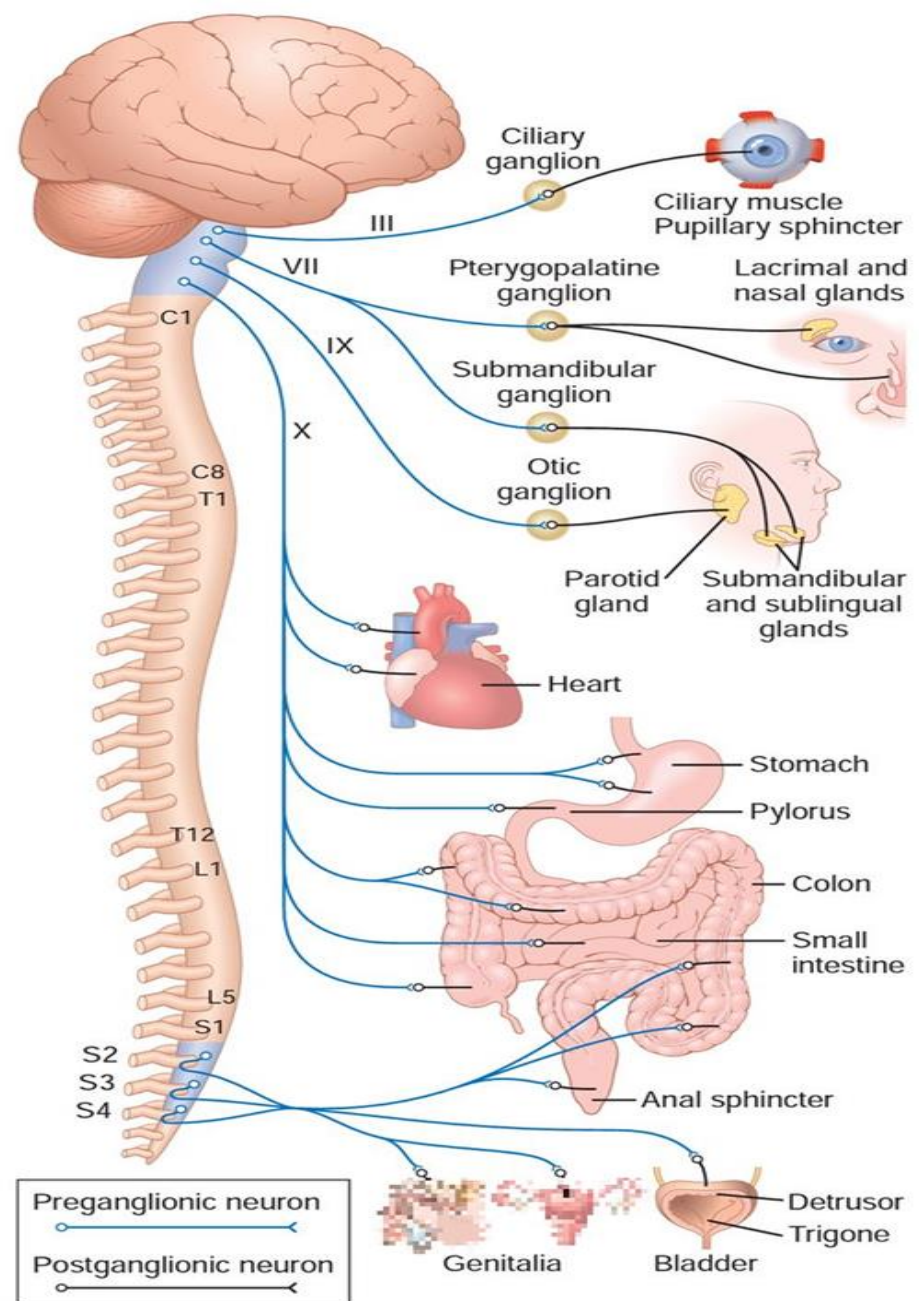
Sympathetic Nerve Fibers in the Skeletal Nerves.

- ❖ Some of the postganglionic fibers pass back from the sympathetic chain into the spinal nerves through *gray rami* at all levels of the spinal cord.
- ❖ These sympathetic fibers are all very small type C fibers, and they extend to all parts of the body by way of the skeletal nerves.
- ❖ They control the blood vessels, sweat glands, and piloerector muscles of the hairs in different parts of the body.
- ❖ About 8 percent of the fibers in the average skeletal nerve are sympathetic fibers, a fact that indicates their great importance.



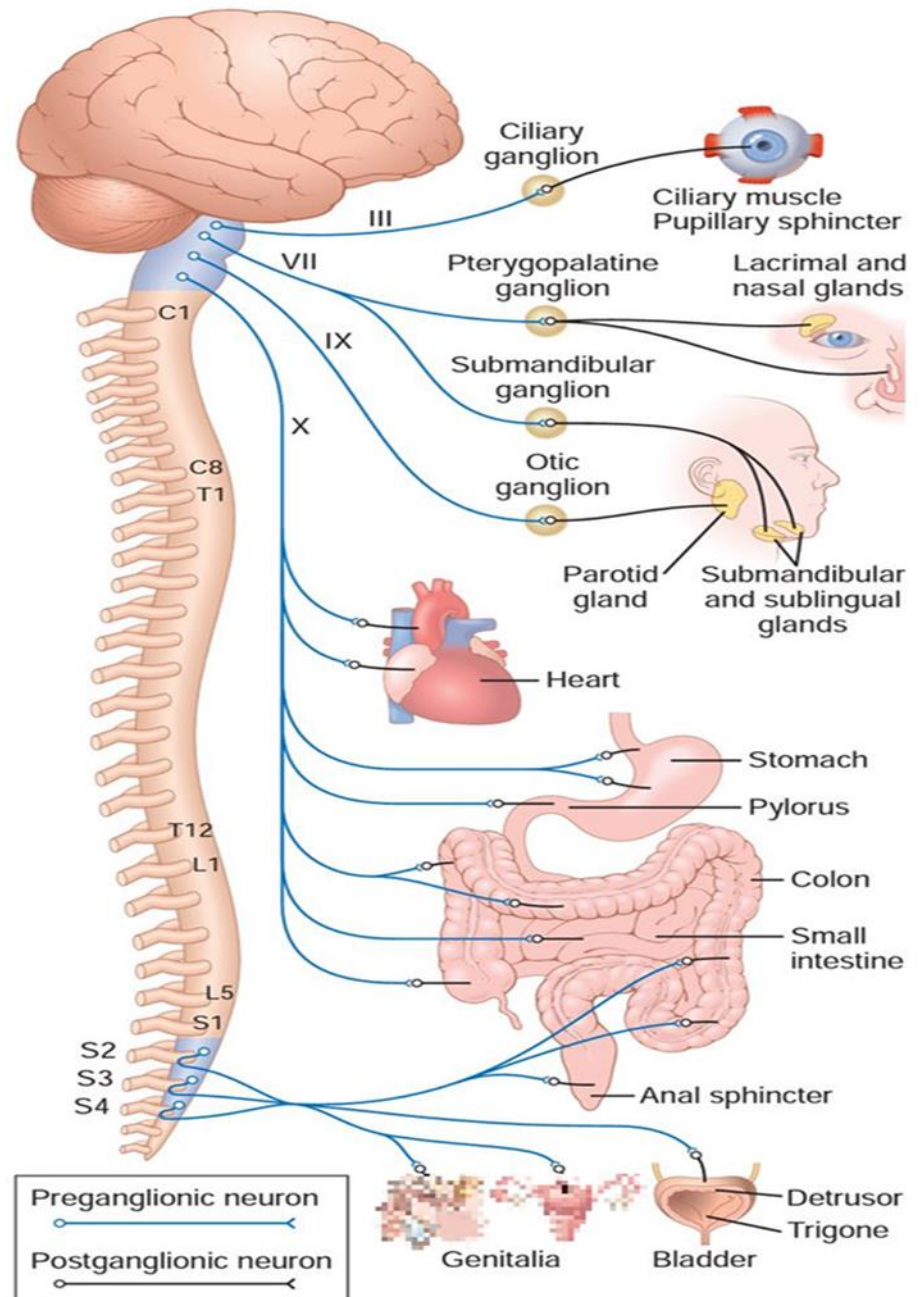
Parasympathetic division (craniosacral division)

- ❖ Preganglionic neurons in the brainstem (nuclei of cranial nerves III, VII, IX, X) and sacral segments of spinal cord (S2-S4)
- ❖ **Oculomotor nerve (III)** : eye ciliary ganglia
- ❖ **Facial nerve (VII)** : salivary gland and anterior 2/3 of tongue, submandibular ganglia
- ❖ **Glossopharyngeal nerve (IX)** : Parotid gland and post 1/3 of tongue, otic ganglion
- ❖ **Vagus nerve (X)** : 70% GIT from esophagus to the half of large intestine
- ❖ **Sacral nerves (pelvic nerves.)** GIT from last half of large intestine to anus, bladder, genital tract



Organization and anatomy of the parasympathetic division

- ❖ Preganglionic fibers leave the brain as cranial nerves III, VII, IX, X
- ❖ Cranial nerve X provides 75% of the parasympathetic outflow
- ❖ Ganglionic neurons in peripheral ganglia located near or within target organs :
 - Terminal ganglion
 - Intramural ganglion



A Comparison of Somatic and Autonomic Function

Feature	Autonomic Nervous System	Somatic Nervous System
▪Site of origin	▪Sympathetic: lateral horn of thoracic and lumbar spinal cord ▪Parasympathetic: brain and sacral spinal cord	▪spinal cord for most; those supplying muscles in head originate in brain
▪Number of neurons from CNS to effector organ	▪Two-neuron chain (preganglionic and postganglionic)	▪Single neuron (motor neuron)
▪Organs innervated	▪Cardiac muscle, smooth muscle, most exocrine and some endocrine glands	▪Skeletal muscle
▪Neurotransmitter at effector organs	▪May be ACh (parasympathetic terminals) or NE (sympathetic terminals)	▪Only ACh
▪Type of control	▪Under involuntary control	▪Subject to voluntary control

Autonomic Effects on Various Organs of the Body

Organ	Effect of Sympathetic Stimulation	Effect of Parasympathetic Stimulation
Eye : Pupil Ciliary muscle	Dilated Slight relaxation (far vision)	Constricted (near vision)
Glands Nasal Lacrimal Parotid Submandibular Gastric Pancreatic	Vasoconstriction and slight secretion	Stimulation of copious secretion (containing many enzymes for enzyme-secreting glands)
Sweat glands	Copious sweating (cholinergic)	Sweating on palms of hands
Blood vessels	Most often constricted	Most often little or no effect
Heart :	Increased rate Increased force of contraction	Slowed rate Decreased force of contraction (especially of atria)

Autonomic Effects on Various Organs of the Body

Organ	Effect of Sympathetic Stimulation	Effect of Parasympathetic Stimulation
Lungs : Bronchi Blood vessels	Dilated Mildly constricted	Constricted ? Dilated
Gut Lumen Sphincter	Decreased peristalsis and tone Increased tone (most times)	Increased peristalsis and tone Relaxed (most times)
Bladder Detrusor Trigone	Relaxed (slight) Contracted	Contracted Relaxed
Basal metabolism	Increased up to 100%	None
Skeletal muscle	Increased glycogenolysis Increased strength	None